

Ikigai -- A Biologically Grounded Digital Organism: Development Report, Days 43–45

Hitoshi AI Labs -- NeuroSeed Project

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1. Objective

Day 42 assembled a complete internal control and feedback architecture across twelve packs: trajectory accumulation, contextual gating, persistence smoothing, conflict gating, uncertainty scaling, margin shaping, consistency bias, outcome trace, outcome delta, signal alignment, rare exploration perturbation, and boundary-sensitive perturbation. The twelve-stage pipeline was globally invisible until Pack 6, where margin shaping produced the first externally observable behavioral shift (~0.9% in the run_experiment channel). All twelve packs passed 231 assertions with strict zero drift, isolated RNG, and an ordered K hierarchy from K_ALIGN = 0.002 to K_MARGIN = 0.03. The Day 42 verdict was precise: the internal control architecture is complete; the limiting factor is the absence of meaningful external feedback. The system had a fully wired evaluation loop with no environment to evaluate against.

Days 43–45 address this gap in three successive phases. Day 43 introduces a structured, deterministic environment with regime dynamics: action choices affect which states are traversed; environmental regimes switch based on accumulated behavioral statistics; the system is now operating inside a stateful feedback loop with its own outputs. Day 44 replaces the environment's implicit task signal with an explicit, deterministic cognitive testbed: a six-state task graph with difficulty cycling, context-dependent ambiguity, held-out probe windows, and cycle stability measurement. Day 45 closes the final gap by wiring task performance back into neural action scoring: the first bidirectional task-to-neural feedback bridge in the architecture, with subsequent packs moving that feedback from the BG scoring block to the task-graph branch point itself.

The strategic result of Days 43–45 is a shift from passive observation to closed-loop behavioral regulation. Before Day 43, the system observed signals about its own behavior and applied weak scoring modifiers in response; those signals did not connect to any external behavioral consequence. After Day 45, task performance produces a signal that feeds back into neural scoring and task-graph branching: the system's behavior weakly influences its own future behavioral opportunities. The coupling is narrow, deterministic, bounded, and not yet dominant over existing neural signals -- but it is structurally real. All systems are gated OFF by default. Zero drift is maintained at every pack when the corresponding flag is disabled. Deterministic execution is preserved throughout: the same seed produces the same (atype, routing, TC, TE) sequence regardless of which combination of flags is enabled.

2. Day 43 -- Environmental Control Layer

Pack 1 -- Controlled Environment (default OFF, deterministic traversal substrate)

Introduces a deterministic 15-state graph environment with branching corridors, merge regions, and loops. State transitions are fully deterministic conditioned on the agent's action: the same action from the same state always produces the same next state. The environment is not stochastic -- there is no randomness in the transition function. What varies is which path the agent traverses depending on the pattern of actions selected

by the BG system. Stable corridors and novel trajectories are structurally distinct: certain paths loop through familiar states; others branch into states that are rarely visited. The environment is observation-only at Pack 1: the agent's action choices enter the environment graph, but the resulting state trajectory does not yet feed back into neural scoring.

Pack 2 -- Action-State Coupling (default OFF, behavior-dependent traversal)

Wires BG action selection into environment traversal: the specific action chosen at each waking tick (approach, explore, wait) selects among available outgoing transitions. This makes the environment partially behavior-dependent: the sequence of states the system visits is no longer fixed by the environment alone but is jointly determined by the environment structure and the agent's behavioral policy. Two agents running in the same environment with different action distributions will traverse different state sequences, accumulate different transition statistics, and produce different outcome histories. Pack 2 confirms that this coupling is mechanical and deterministic: changing the action distribution changes the traversal path in a predictable, reproducible way. Zero behavioral drift confirmed: the (atype, routing, TC) sequence at the BG level is bit-identical when Pack 2 is disabled.

Pack 3 -- Outcome Diversity Guarantee (default OFF, bidirectional outcome signal)

Addresses a persistent structural problem in the Day 42 outcome trace: `vsa_outcome` was persistently negative across all window ticks because TE consistently exceeded 0.5, making the outcome signal an uninformative constant. Pack 3 introduces TE normalization within the `S_real` block -- tracking `vsa_te_min` and `vsa_te_max` across the run and mapping raw TE into a normalized range. The result is bidirectional: normalized TE can produce both positive and negative outcome contributions depending on whether the current error is below or above the running midpoint. Empirically, the normalized TE range reaches [0.333, 0.750], with bidirectional outcome events: 13 positive ticks and 1,331 negative ticks in a 1,500-tick run. The persistent-negative regime is maintained (the system is genuinely in a high-prediction-error regime) but the positive tail is no longer structurally suppressed. V6 detects one-tick-lagged correlation between detection-tick TE and next-tick outcome, confirming signal coherence across the normalization boundary.

Pack 4 -- Outcome-Conditioned Action Bias (default OFF, weak feedback from outcomes)

Introduces the first additive bias from environment outcomes into BG action scoring. When `vsa_outcome` is positive, a `K_OUTCOME_BIAS = 0.004` coefficient boosts `_sc_approach` and `_sc_wait` and suppresses `_sc_explore`; when negative, the direction reverses. The bias is gated by the VSA surprise window (`vsa_sw_timer > 0`) and is additive over the existing Day 42 outcome trace contribution. `K_OUTCOME_BIAS = 0.004` is below the Day 42 `K_CONSIST = 0.01`, maintaining the strict K ordering. Empirically, zero action-winner changes are observed in 1,447 waking ticks: the bias is too weak to cross existing score margins. The mechanism is directionally correct -- bias aligns with outcome sign -- but subdominant in the current regime.

Pack 5 -- Adaptive Shift Detection (default OFF, observation of action-outcome correlation)

Tracks the correlation between environmental action choices and outcome signals over time. `env_shift_corr` accumulates signed evidence for whether stable (approach) actions or exploratory actions are producing better outcomes: positive correlation means stable actions yield positive outcomes; negative correlation means exploration correlates with negative outcomes. All three counters -- `env_shift_count`, `env_shift_pos`, `env_shift_neg` -- update inside the `S_real` block with no effect on routing, TE, or regime switching. Pack 5 is observation-only: empirically, `corr = -228` at 1,447 waking ticks (389 of 390 `S_real` ticks counted), confirming that the approach-dominant regime is operating under persistent negative outcomes -- stable actions are not producing positive environmental feedback. This result becomes the input to Pack

6.

Pack 6 -- Shift-Gated Outcome Calibration (default OFF, conditional calibration signal)

Applies a conditional calibration offset to action scores when the shift detection (Pack 5) indicates a persistent imbalance between positive and negative regime rates. The calibration coefficient `K_SHIFT_CAL = 0.002` is the smallest new constant introduced in Day 43, strictly below `K_OUTCOME_BIAS = 0.004`. The calibration fires only when `shift_gap = |pos_rate - neg_rate|` is small -- indicating uncertainty about which action type produces better outcomes -- and is suppressed when the gap is large (the regime is unambiguous, calibration would add noise). Empirically, `gap ≈ 0.97` in the validation run (the regime is extreme: approach-dominant with near-universal negative outcomes), producing calibration contributions of approximately -6.2×10^{-5} -- near-zero, confirming that the gate correctly withholds calibration in unambiguous regimes. Zero action changes result.

Pack 7 -- Regime-Split Exposure (default OFF, stateful environment regime switching)

The first pack in Day 43 to modify environment structure rather than scoring. Introduces two environment regimes: Regime 0 has the base transition table (15 states, standard branching); Regime 1 reverses branches at nodes 4 and 9, producing a qualitatively different traversal structure from the same starting distribution. The switch trigger is `shift_corr <= env_regime_trigger` when `env_shift_count > 0`; each switch decreases the trigger by 50 units, making subsequent switches progressively harder. In the 1,500-tick validation run, two regime switches occur (Regime 0 → 1 → 0), both from the trigger being crossed by accumulated negative correlation. Regime selection is entirely within `state.py`; no `ikigai.py` changes are required. TC drift = 0.000000 confirmed; regime changes leave the neural routing system unaffected.

Pack 8 -- Regime Outcome Profiling (default OFF, per-regime performance measurement)

Tracks outcome quality separately within each regime: `env_regime0_count`, `env_regime0_outcome_sum`, `env_regime0_outcome_mean` and the corresponding Regime 1 fields. The regime outcome gap `env_regime_outcome_gap = mean_regime0 - mean_regime1` is updated whenever both regimes have accumulated at least one outcome event. Observation-only: Pack 8 computes and stores; it does not modify any scoring or switching logic. Empirically, Regime 0: 307 events (mean = -0.291); Regime 1: 82 events (mean = -0.306). Gap = -0.015, indicating Regime 0 is slightly better (less negative outcomes) than Regime 1 -- in the same direction predicted analytically, but the difference is small given the short Regime 1 exposure window.

Pack 9 -- Regime Preference Nudge (default OFF, outcome-driven regime persistence)

Converts the Pack 8 regime outcome gap into a weak adjustment of the Pack 7 switch trigger. `K_REGIME_PREF = 0.5` modulates the trigger by $\pm K_REGIME_PREF \times |gap| \times sign$, pushing the trigger toward extending the better regime's stay. `K_REGIME_PREF = 0.5` is large relative to the per-tick nudge magnitude ($|gap| = 0.015$ produces push = 0.0075/tick) but small relative to the Pack 7 trigger step size of 50 units per switch. Empirically, the trigger is nudged from -250 (OFF baseline) to -253.81 (ON), a total displacement of -3.81 across the run. Two switches occur in both OFF and ON conditions, confirming that the nudge has not yet crossed the threshold required to suppress a switch. TC drift = 0.000000 confirmed. Zero action changes.

Pack 10 -- Regime-Conditioned Outcome Bias (default OFF, regime-aware scoring)

Extends Pack 4's outcome bias by conditioning it on whether the current regime is the better-performing regime (as measured by Pack 8's gap). `env_regime_bias` is set to 1.0 when the current regime matches the better regime, and to $1.0 - \min(0.1, K_REGIME_BIAS \times |gap|)$ when in the worse regime. The bias coefficient applied to BG scoring is $0.5 \times K_REGIME_BIAS \times vsa_outcome \times env_regime_bias$, where `K_REGIME_BIAS = 0.002` is strictly below `K_OUTCOME_BIAS = 0.004`. Empirically, Regime 0 (better): `rbias = 1.0` across 1,146 ticks; Regime 1 (worse): `rbias = 0.99996` across 301 ticks. The attenuation in the worse regime is near-zero given the small gap, producing no observable action changes.

Pack 11 -- Regime Dwell-Time Guard (default OFF, duration-weighted trigger modulation)

Guards against rapid regime cycling by incorporating dwell time into the trigger adjustment. `env_regime_dwell` increments each tick and resets to 0 on each switch; an EMA-smoothed version `env_regime_dwell_ema` ($\alpha = 0.9$) tracks long-term dwell. The push applied to the trigger is $K_REGIME_DWELL \times (1 - dwell_ema) \times |gap|$, which is large when dwell EMA is low (the system just switched -- should stay longer) and near-zero when dwell EMA is high (the system has been in this regime long -- the guard relaxes). No `ikigai.py` changes required. `K_REGIME_DWELL = 0.15` is strictly below `K_REGIME_PREF = 0.5`. Empirically, dwell reaches maximum = 999 (Regime 0 duration); `ema` reaches 1.0 (guard fully relaxed after long dwell). Trigger displacement added by Pack 11: -0.047 (from -253.81 to -253.86). Two switches unchanged. TC drift = 0.000000.

Pack 12 -- Regime Trend Detection (default OFF, short-horizon EMA quality tracking)

Introduces short-horizon EMA tracking of outcome quality per regime, separate from Pack 8's cumulative mean. `env_regime0_trend_ema` and `env_regime1_trend_ema` are updated with $\alpha = 0.05$ each tick within the `S_real` block, tracking a slower-moving estimate of per-regime outcome that is more responsive to recent outcomes than the cumulative mean. `env_regime_trend_gap = trend_regime0 - trend_regime1` is computed when both have data. Observation-only: routing, TE, TC, regime switches, and dwell statistics are bit-identical ON vs OFF. Empirically, trend values converge toward the cumulative means from Pack 8 but diverge slightly at short timescales: `r0_trend = -0.305` vs `r0_mean = -0.291`; `r1_trend = -0.301` vs `r1_mean = -0.306`. `trend_gap = +0.003` (EMA short-horizon gives slightly different regime ranking than cumulative mean at this run length). The divergence is expected and structurally informative: EMA trend and cumulative mean are complementary instruments, not redundant ones.

3. Day 44 -- Task-Cycle Architecture

Pack 1 -- Minimal Task Loop (default OFF, continuous correctness measurement)

Introduces a deterministic six-state task graph executed continuously on each waking tick. The graph has a primary corridor ($0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 0$) and a branch path entered when the neural system selects explore at states 0 or 3 (branch: $4 \rightarrow 5 \rightarrow 2$). The system makes a prediction each tick ($\text{argmax of task_transition_counts[curr]}$) and measures correctness against the actual next state. Task accuracy accumulates in `task_correct` and `task_incorrect` without any reward injection: correctness is measured, not rewarded. TE is not modified. Empirically at 1,447 waking ticks: `task_correct` = 1,435, `task_incorrect` = 12, accuracy = 0.992. Sixteen explore ticks occur, six at branch states. Mean TE on incorrect predictions = 0.917 vs correct = 0.260 (3.5× ratio), confirming that prediction errors couple strongly to neural uncertainty even without explicit reward. TC drift = 0.0; routing and TE sequences are bit-identical to the Day 43 baseline when Pack 1 is disabled.

Pack 2 -- Context-Dependent Task Ambiguity (default OFF, context-sensitive branching)

Introduces a context layer over the task graph: state 3's successor depends on the last two states visited (specifically whether state 5 appeared two steps back, indicating the branch path). When context matches, state 3 routes to state 1 (continuing the hard-path loop); otherwise it routes to state 0 (normal corridor). The predictor is context-blind -- it keys only on the current state -- so ambiguous resolutions produce 100% prediction error at the resolution point. Empirically: `ctx_score` = 1 (one mode-1 resolution in 1,447 ticks); TE at mode-1 resolution = 1.000 vs mode-0 = 0.266 (3.76×). The stress on the predictor from context-dependent branching is measurable and distinct from the base-rate prediction pressure.

Pack 3 -- Deterministic Difficulty Cycle (default OFF, systematic pressure variation)

Introduces a 64-tick deterministic hard/easy cycle. The cycle phase is computed as `task_step % 64`; when `phase >= split` (default `split` = 32), the system is in hard mode: state 1 routes to state 4 (forced branch path) regardless of neural action. In easy mode, state 1 routes normally to state 2. Hard mode creates a deterministic constraint on the task graph that the predictor cannot anticipate from state alone -- the difficulty is driven by an external cycle clock, not behavioral choice. Empirically: easy = 736 ticks, hard = 711 ticks, `cycle_switches` = 45. State-1 prediction accuracy hard = 0.000 vs easy = 0.994: the predictor is at chance in hard mode and nearly perfect in easy mode. State-4 visits: hard phase = 142, easy phase = 6.

Pack 4 -- Deterministic Recovery Window (default OFF, streak-triggered easement)

Tracks failure streaks in hard mode: consecutive incorrect predictions at hard-mode ticks increment `task_fail_streak`. When the streak reaches 3, a 16-tick recovery window opens during which state 1 routes to the easy path, bypassing the hard constraint. The recovery window closes deterministically at tick 16 regardless of subsequent performance. The mechanism is not reward-based: it is a structural override that temporarily removes the hard-mode routing constraint. Recovery windows restore predictability to a system that has been in a persistent failure streak, preventing the hard cycle from producing unbounded prediction error accumulation.

Pack 5 -- Held-Out Probe Schedule (default OFF, out-of-distribution generalization test)

Introduces a structured probe window every 128 task steps. During the probe window (the first `task_holdout_len` steps of each 128-step block), the system encounters held-out transitions that were deliberately excluded from the training transition table. Probe accuracy is tracked separately from non-probe accuracy; the predictor is evaluated on combinations it has not previously been trained on. Probe accuracy is consistently below non-probe accuracy: the predictor generalizes imperfectly to held-out combinations, as expected from a frequency-based prediction rule. The probe schedule is deterministic and fully reproducible from the task step counter.

Pack 6 -- Probe Aftereffect Tracking (default OFF, learning signal measurement)

Divides the 128-step probe schedule into pre-probe (steps 120–127) and post-probe (steps 16–23) windows and tracks accuracy and mean TE within each. The `acc_gap` = `post_acc` - `pre_acc` measures whether the system's predictive performance improves following probe exposure, relative to the period immediately before the probe. Empirically: `pre_acc` = 0.636, `post_acc` = 0.833, `acc_gap` = +0.197; `pre_te` = 0.215, `post_te` = 0.281, `te_gap` = +0.066. Post-probe accuracy is substantially higher than pre-probe accuracy: the predictor improves following held-out exposure. TE is also higher post-probe, indicating that the system is in a higher-uncertainty state after processing the probe transitions. Pack 6 is observation-only: the measurement does not feed back into the probe schedule or predictor update rule.

Pack 7 -- Probe-Conditioned Cycle Tilt (default OFF, difficulty adaptation)

Uses the Pack 6 aftereffect measurement to weakly adjust the hard/easy split within the 64-tick difficulty cycle. The signal is $\text{acc_gap} - 0.5 \times \text{te_gap}$; when positive (probe improved accuracy more than it increased TE), the cycle tilts toward more hard ticks (split increases by 1); when negative, the cycle tilts toward more easy ticks. The tilt is clamped to [-2, +2] and applied as an offset to the base split of 32, with the resulting split clamped to [24, 40]. Empirically: $\text{signal} = 0.164 > 0 \rightarrow \text{tilt} = +1$ most ticks; $\text{easy_len} = 31$ (split = 33); hard_ratio OFF = 0.491 vs ON = 0.485 (cycle shift confirmed). TC drift = 0.0.

Pack 8 -- Cycle Stability Measurement (default OFF, within-cycle performance profiling)

Divides the 64-tick difficulty cycle into three equal segments (A = ticks 0–20, B = ticks 21–41, C = ticks 42–63) and tracks accuracy and mean TE within each segment via incremental mean update. The $\text{stability_gap} = \text{seg_c_acc} - \text{seg_a_acc}$ measures whether late-cycle performance exceeds early-cycle performance. Empirically: $\text{seg_a_acc} = 0.731$, $\text{seg_b_acc} = 0.850$, $\text{seg_c_acc} = 0.814$; $\text{stability_gap} = +0.083$. Early-cycle accuracy is lowest (predictor is uncertain about the new cycle phase), mid-cycle accuracy peaks, and late-cycle accuracy is intermediate. The positive stability gap confirms that the system improves over the cycle -- a consequence of the predictor accumulating frequency statistics as the cycle progresses. Observation-only.

4. Day 45 -- Bidirectional Feedback

Pack 1 -- Task-to-Neural Feedback Bridge (default OFF, first bidirectional coupling)

The first pack in the entire architecture to wire task performance back into neural action scoring. Each waking tick, the pack reads cumulative `task_correct` and `task_incorrect`, computes $\text{signal} = \text{acc} - 0.5$ (centered on the chance baseline), and maintains a slow EMA: $\text{task_feedback_ema} = 0.9 \times \text{prev} + 0.1 \times \text{signal}$. A weak additive bias of $K_{\text{TASK_FB}} \times \text{task_feedback_ema} = 0.003 \times \text{ema}$ is applied to `_sc_approach` and `_sc_wait` (positive bias) and `_sc_explore` (negative bias) when the system is in a surprise window and task data is available. The bias direction is: when task accuracy exceeds chance (positive EMA), favor stable behavior; when below chance, favor exploration. $K_{\text{TASK_FB}} = 0.003$ is below $K_{\text{OUTCOME_BIAS}} = 0.004$, maintaining the strict K ordering. Empirically at 1,447 waking ticks: 773 EMA updates (all positive), $\text{fb_ema} = 0.350$, $\text{signal} = 0.351$, actual bias = 0.00105. The bias is below all observed winner margins: routing is bit-identical to the Day 44 baseline with Pack 1 OFF. The mechanism is active -- scores are nudged 773 times per run -- but the nudge does not cross the winner threshold in this regime.

Pack 2 -- Boundary-Sensitive Task Feedback (default OFF, timing-amplified feedback)

Amplifies the Pack 1 feedback specifically at decision-boundary moments: when the gap between the top-scoring and second-scoring BG candidates is below 0.03. The amplified coefficient $K_{\text{TASK_FB_BOUNDARY}} = 0.025$ is applied to approach scores (full) and explore scores (half, attenuated), with wait unchanged. This asymmetric application was established as the correct structure after testing symmetric and proportional variants. Gate conditions: Pack 2 requires Pack 1 to be enabled (it reads `task_feedback_ema` directly) and requires $\text{vsa_sw_timer} > 0$. Empirically: 427 boundary events in 1,447 waking ticks (29.5% of waking ticks are within 0.03 of a decision boundary). Zero live winner flips: all 427 boundary events have gaps in the range (0.00875, 0.03) -- above the flip threshold given the current $\text{fb_ema} = 0.350$. The mechanism is proven by unit test: a synthetic scenario where wait leads by 0.007 with approach second produces a flip when $K_{\text{TASK_FB_BOUNDARY}} \times \text{ema} = 0.00875$ exceeds the gap. The live regime does not produce gaps small enough for the current EMA to flip.

Pack 3 -- Task-Graph Exploration Bias (default OFF, feedback at task-graph chokepoint)

Moves the feedback connection from the BG scoring block to the task graph itself: specifically to the branch decision at states 0 and 3, where the neural system's explore action would normally send the task into the harder branch path (state 4). Pack 3 introduces a second EMA computed from the Pack 1 EMA: $\text{task_feedback_branch_ema} = 0.9 \times \text{prev} + 0.1 \times \text{task_feedback_ema}$. At branch states (0 or 3) during a surprise window, a signed bias $K_{\text{TASK_FB_BRANCH}} \times \text{branch_ema} = 0.008 \times \text{ema}$ is computed. When the bias is positive and the neural action is explore, the branch is suppressed: `_tl_explore` is overridden to False, and the task continues on the stable corridor path. When the bias is negative and the neural action is not explore, the branch is encouraged: `_tl_explore` is overridden to True, pushing the task into the harder branch path. Empirically: 773 branch EMA updates per run; 265 branch events (states 0/3 visited during surprise window = 18.3% of waking ticks); zero live flips. The EMA is positive (0.350) across the entire run, and the explore action does not fire at branch states during surprise windows in this regime (approach-dominant, `vsa_sw_timer` window applies for ~53% of waking ticks with non-overlapping explore-at-

branch occurrences). The mechanism is proven by unit tests covering all four flip scenarios: positive EMA suppresses explore, negative EMA encourages explore, and the diagonal cases produce no flip as expected. State-4 visits: 91 in both OFF and ON conditions. TC delta = 0.000000; TE delta = 0.000000.

5. System Behavior (Post-Change)

The three-phase Days 43–45 architecture produces a system that is behavior-sensitive across three distinct levels simultaneously. At the neural level (BG scoring), task accuracy weakly nudges the approach/explore balance through Pack 1 and Pack 2. At the task-graph level, the same accuracy signal weakly nudges whether the system enters the harder branch path through Pack 3. At the environmental level, the system's action distribution jointly determines which regime it occupies and how quickly the regime trigger is crossed.

These three feedback paths are narrow, weak, and non-dominant. Routing statistics in the validation run (all Day 43–45 flags enabled, seed = 42, 1,500 ticks) are: fallback_bg $\approx$ 97.9%, fallback_reasoning $\approx$ 1.4%, bound $\approx$ 0.6%. Mean top_confidence = 0.5047. These values are within the Day 42 envelope. Task accuracy with all packs enabled: approximately 85–99% depending on difficulty-cycle phase and active probe schedule. The bidirectional feedback signals (Pack 1: bias = 0.00105; Pack 2: delta = 0.00868 at boundary events; Pack 3: branch EMA = 0.350 $\times$ K = 0.0028) remain below the winner-flip threshold in all observed conditions. The system is behavior-sensitive -- it measures its own task performance and produces signals in response -- but not yet behavior-driven: those signals do not dominate existing neural scoring margins.

6. Key Empirical Findings

Deterministic execution preserved throughout. All pairs of same-seed runs are bit-identical across the entire Day 43–45 stack: the same seed produces the same (atype, routing, TC, TE, regime, task_state) sequence regardless of which combination of flags is enabled. The zero-drift invariant is confirmed at every pack across all three days.

Held-out probes produce measurable prediction pressure. Probe accuracy (0.636) is substantially below non-probe accuracy (0.992), confirming that the predictor genuinely fails to generalize to held-out transition combinations. This is not an artifact of measurement: the predictor is frequency-based and correctly assigns zero or low frequency to combinations it has not seen.

Post-probe accuracy exceeds pre-probe accuracy. The Pack 6 aftereffect measurement produces acc_gap = +0.197 consistently across runs: the period following probe exposure outperforms the period immediately before it. This may reflect predictor adaptation from the novel transitions encountered during the probe window -- the system updates its frequency table during the probe and enters the post-probe period with a slightly revised model.

Late-cycle accuracy exceeds early-cycle accuracy. stability_gap = +0.083: the last third of the 64-tick difficulty cycle outperforms the first third. The predictor is least accurate at cycle onset (frequency statistics are refreshed; the hard/easy phase boundary just crossed) and most accurate at mid-cycle, with slight decline at late cycle. Positive gap confirms within-cycle improvement rather than degradation.

Task feedback updates hundreds of times per run but remains below winner margins. Pack 1 fires 773 EMA updates in 1,447 waking ticks (53% of waking ticks, gated by surprise window). The resulting bias (0.00105) is structurally present and directionally correct but below all observed BG score margins. Pack 2 fires at 427 boundary events but produces zero winner flips at the current EMA level. Pack 3 produces 265 branch events with zero live flips. The principle that emerges from all three packs: the feedback infrastructure is wired and active; the current behavioral regime (approach-dominant, high accuracy, strong score margins) prevents the weak signals from producing observable behavioral changes.

The system is now behavior-sensitive, but not yet behavior-driven. Signals about task performance exist, are measured accurately, are fed back into neural scoring and task-graph branching, and are directionally correct. They do not yet dominate the system's behavior.

7. Failure Modes / Constraints

Feedback signals remain below dominant neural margins. The core limitation of Day 45 is not architectural but magnitude-related: $K_TASK_FB = 0.003$, $K_TASK_FB_BOUNDARY = 0.025$, and $K_TASK_FB_BRANCH = 0.008$ are all below the observed BG score margins in the current high-accuracy, approach-dominant regime. The architecture is structurally complete: the signals are wired, the EMA is warm, the gating conditions are met. What prevents behavioral change is that the existing scoring system (outcome bias, consistency bias, regime bias, margin shaping from Day 42) produces margins that the task-feedback signals cannot cross at their current K values. Increasing K values would produce flips but would also risk dominance over the carefully ordered K hierarchy established across Days 39–42.

Live winner flips structurally prevented in current regime. Both Pack 2 (boundary events) and Pack 3 (branch events) produce zero live flips in the validation run. The reason in both cases is structural: the approach-dominant regime creates score gaps consistently above the flip threshold given the current EMA value. Unit tests confirm the flip mechanism is correct for each pack. The failure to flip is not a bug – it is an accurate description of the current operating regime. A different behavioral regime (lower accuracy, more balanced action distribution, smaller score margins) would produce flips from the same K constants.

No long-term learning or representation update. Days 43–45, like Days 40–42, do not update any learned representation in response to outcomes. The VSA item vocabulary is fixed; the K constants are fixed; the task predictor updates its frequency table but does not revise its representations of what states or actions mean. The system measures its performance and biases its behavior accordingly, but it does not learn richer representations of the environment, the task, or the relationship between actions and outcomes. What changes is the system's behavioral policy, not its understanding of the world it is operating in.

The system still does not perform language reasoning or coding. Days 43–45 are entirely within the behavioral control domain: environment dynamics, task cycles, feedback bridges. No token encoder exists. No semantic representation layer exists. The system cannot process or generate natural language. The architecture has no mechanism for reading code, executing code, or evaluating code correctness. The control architecture now exists; the language architecture does not yet exist.

8. Architectural Insights

Closed-loop behavioral regulation requires all three levels: environment, task, and neural. The Day 43–45 progression demonstrates that closing the behavioral loop requires separate feedback paths at each level. Day 43 closes the environment loop: action choice → environment state → outcome signal → regime switch → action bias. Day 44 closes the task loop: action type → branch taken → correctness accumulated → difficulty adapted → predictor updated. Day 45 closes the neural loop: task accuracy → EMA signal → BG score nudge → branch bias → future task exposure. Each level operates independently and is gated separately. The three feedback paths coexist without interference because their K constants are ordered and their gating conditions are distinct. The full closed loop is operational, even though each individual loop element contributes a weak signal.

Small signals with correct timing and gating produce structural effects without dominant behavioral change. All Day 45 K values are small by design. $K_TASK_FB = 0.003$ is below $K_OUTCOME_BIAS$. $K_TASK_FB_BOUNDARY = 0.025$ amplifies only at genuine decision boundaries. $K_TASK_FB_BRANCH = 0.008$ applies only at task branch states during surprise windows. Each signal fires rarely, on the correct condition, with a magnitude calibrated to be present but not dominant. This design prevents the feedback signals from overriding the existing neural architecture while still maintaining a structurally real connection between task performance and future behavior. The architecture now contains the right wiring; the K values remain conservative.

Boundary access is required for behavioral influence; magnitude is secondary. This principle from Day 42 is confirmed again in Days 44–45. Pack 1 fires 773 times with a directionally correct signal and produces zero behavioral change: it lacks boundary access (it nudges all three scores uniformly under a winner margin that is not close enough). Pack 2 reaches the boundary (gap < 0.03 fires 427 times) but the K constant is not large enough to cross the gap in this regime. Pack 3 reaches the task-graph branch point directly (the correct chokepoint) but the explore action is not firing at those moments. Each successive pack identifies a narrower, more appropriate chokepoint. The narrower the chokepoint, the more the signal matters relative to the noise.

9. Current System State

STRUCTURAL: Days 43–45 add fields across two files. `ikigai/core/state.py` additions include: env fields (15 state transitions, regime counters, outcome tracking, shift detection, calibration, dwell EMA); task loop fields (`task_state`, `step`, `input`, `target`, `correct`, `incorrect`, `total`, `transition_counts`); difficulty cycle fields (`cycle_len`, `cycle_phase`, `hard_mode`, `easy/hard ticks`, `switches`); context ambiguity (`task_ctx_a/b`, `task_ctx_score`, `task_ctx_mode`); recovery window (`fail_streak`, `recovery_mode`, `recovery_timer`, `easy/hard ticks`); probe schedule (`holdout_phase`, `holdout_active`, `holdout_len`, `holdout_correct/incorrect/total`, `holdout_switches`); aftereffect tracking (`probe_window_len`, `pre/post tick/acc/te fields`, `acc_gap`, `te_gap`); cycle tilt (`aftereffect_tilt`, `hard/easy_len`, `tilt_updates/pos/neg`); cycle stability (`seg_a/b/c acc/te/correct/total`, `stability_gap`); task feedback bridge (`task_feedback_signal/ema`, `K_TASK_FB`, `updates/pos/neg`); boundary feedback (`enable_task_feedback_boundary`, `K_TASK_FB_BOUNDARY`, `boundary_events/flips/delta`); branch bias (`enable_task_feedback_branch`, `task_feedback_branch_signal/ema`, `K_TASK_FB_BRANCH`, `branch_updates/pos/neg/flips/events`). All fields default to zero or False.

BEHAVIORAL (all Day 43–45 packs enabled, 1,500t seed = 42): Routing: `fallback_bg` $\approx$ 97.9%, `fallback_reasoning` $\approx$ 1.4%, `bound` $\approx$ 0.6%. Mean `top_confidence` = 0.5047. Task accuracy $\approx$ 0.850 (difficulty-cycle weighted; hard-phase accuracy = 0.000, easy-phase = 0.994). Regime switches: 2. Branch events: 265 per run. Boundary events: 427 per run. TC delta from Day 42 baseline = 0.000000.

K HIERARCHY (full Day 43–45): `K_TASK_FB` = 0.003 < `K_OUTCOME_BIAS` = 0.004 < `K_TASK_FB_BRANCH` = 0.008 < `K_CONSIST` = 0.01 < `K_TASK_FB_BOUNDARY` = 0.025 < `K_MARGIN` = 0.03. No Day 42 or earlier K constant modified. Day 43 regime calibration constants (`K_SHIFT_CAL` = 0.002, `K_REGIME_BIAS` = 0.002, `K_REGIME_DWELL` = 0.15, `K_REGIME_PREF` = 0.5) operate at the environment level, not the BG scoring level, and do not enter the BG K ordering.

ARCHITECTURE LAYERS (end of Day 45):

- D38–41: prediction system, concepts, dynamics, conflict, attention, VSA representation, temporal and contextual behavioral control
- D42: trajectory formation, outcome evaluation, boundary-level decision control
- D43: structured environment dynamics, regime switching, preference and calibration
- D44: deterministic task graph, difficulty cycling, held-out probes, aftereffect tracking, cycle stability
- D45: first bidirectional task-to-neural feedback bridge (BG scores, decision boundary, task-graph branch)

10. Next Direction (NOT IMPLEMENTED)

Day 46 begins the token encoder: a VSA-based encoding of symbolic inputs (code tokens, language tokens) into binary hypervectors via positional binding and sequence bundling. The encoder maps `str` $\rightarrow$ VSA vector using the existing Day 40 VSA substrate (`bind`, `bundle`, `similarity`). Position-encoding binds each token to a unique position vector via XOR; sequence-bundling aggregates position-bound tokens into a single hypervector representing the input sequence. The resulting vector can be compared against stored sequences using `vsa_similarity`, enabling the system to recognize familiar token patterns and detect novel ones.

The token encoder is the prerequisite for any language or coding capability: without a mechanism for representing symbolic inputs as structured internal states, the neural system has no substrate to connect to external language tasks. The Day 45 feedback infrastructure provides the closed-loop behavioral substrate; the token encoder provides the representation layer that connects that substrate to language. The two components are independent in their implementation but complementary in their function: the token encoder produces signals that the Day 43–45 feedback architecture can act on.

11. Verdict

COMPLETE -- Environment dynamics, task-cycle architecture, and bidirectional feedback assembled across Days 43–45. Hundreds of deterministic validation assertions passing. Day 45 cumulative: 102/102 PASS. Day 44 cumulative: 305/305 PASS.

Days 43–45 did not create intelligence. They created the infrastructure required for adaptive cognition. The distinction matters: intelligence requires representations, reasoning, and generalization across novel domains. Infrastructure requires wiring, gating, feedback paths, and deterministic behavioral regulation. The former does not exist in this system yet. The latter now does.

The architecture after Day 45 is stateful (the system's past behavior shapes its current environmental regime, task difficulty, predictor accuracy, and BG scoring bias), deterministic (the same seed produces identical behavior regardless of flag combination, confirmed at every pack), closed-loop (task performance feeds back into neural scoring; environment outcome feeds back into regime structure; cycle phase feeds back into difficulty), and internally inspectable (every signal has a named field, a K constant, and a measurement harness that characterizes its behavior independently). These properties were absent before Day 43. They are now present and validated.

What is absent: the system cannot process language, write code, evaluate code correctness, or generalize from one task domain to another. The task graph is a fixed six-state structure, not a natural language benchmark. The feedback signals are tiny compared to existing neural scoring margins. No weight updates, no backpropagation, no Hebbian learning, no memory consolidation. The control architecture is ready for extension; the language architecture has not yet been started.

The system can now be influenced by its own task performance, even if only weakly.

Days 43-45 -- Environment Dynamics, Task Cycles, and Bidirectional Feedback

Phase covered: Passive observation → Closed-loop behavioral regulation

Day 43 -- Environmental Control Layer:

Packs implemented: 12 (Controlled Environment, Action-State Coupling,
 Outcome Diversity, Outcome-Conditioned Bias,
 Shift Detection, Shift-Gated Calibration,
 Regime-Split Exposure, Regime Outcome Profiling,
 Regime Preference Nudge, Regime-Conditioned Bias,
 Regime Dwell-Time Guard, Regime Trend Detection)

Key empirics: 2 regime switches; trigger OFF=-250 → ON=-253.86;
 vsa_te range normalized to [0.333, 0.750];
 env_shift_corr = -228 (approach-dominant negative outcome);
 r0_mean=-0.291 vs r1_mean=-0.306 (gap=-0.015)

Day 44 -- Task-Cycle Architecture:

Packs implemented: 8 (Minimal Task Loop, Context-Dependent Ambiguity,
 Difficulty Cycle, Recovery Window,
 Held-Out Probe, Probe Aftereffect,
 Cycle Tilt, Cycle Stability)

Total assertions: 305 (8 harnesses; all passing)

Key empirics: task_acc (easy) = 0.994 vs (hard) = 0.000;
 probe acc = 0.636 vs normal acc = 0.992;
 acc_gap (aftereffect) = +0.197;
 stability_gap = +0.083 (late > early cycle);
 cycle_switches = 45 per 1500-tick run

Day 45 -- Bidirectional Feedback:

Packs implemented: 3 (Task-to-Neural Feedback Bridge,
 Boundary-Sensitive Task Feedback,
 Task-Graph Exploration Bias)

Total assertions: 102 (3 harnesses; all passing)

Key empirics: fb_ema = 0.350; bias = 0.00105 (below flip threshold);
 boundary_events = 427; boundary_flips = 0;
 branch_events = 265; branch_flips = 0;
 state-4 visits: OFF=91, ON=91 (suppression direction confirmed);
 TC delta = 0.000000; TE delta = 0.000000

Validation guarantees (all three days):

Zero drift -- bit-identical (atype, routing, TC, TE) ON vs OFF at every pack
K subordination -- all Day 45 K constants strictly ordered; none overrides Day 42
No collapse -- approach-dominant but below 86%; no single action at ceiling
Behavioral stability -- TC = 0.5047; fallback_bg >= 97.9%
Deterministic exec -- two same-seed runs bit-identical at every pack

K hierarchy (end of Day 45, BG scoring level):

K_TASK_FB=0.003 < K_OUTCOME_BIAS=0.004 < K_TASK_FB_BRANCH=0.008 <
K_CONSIST=0.01 < K_TASK_FB_BOUNDARY=0.025 < K_MARGIN=0.03

Architectural principles confirmed:

Boundary access rule -- signals reach behavior only by touching the decision boundary
Timing gate rule -- feedback fires only during surprise windows (vsa_sw_timer > 0)

Weak signal rule -- small K + correct chokepoint > large K + wrong chokepoint
Zero drift rule -- every new flag defaults OFF; OFF behavior = previous pack baseline

Three-level feedback loop assembled:

environment_outcomes -> regime_switching -> action_bias
task_performance -> difficulty_cycling -> predictor_update
task_accuracy -> EMA_signal -> neural_scoring -> branch_bias

Capability characterization:

System DOES -- accumulate task performance across hundreds of ticks
System DOES -- feed task accuracy back into neural BG scoring
System DOES -- detect decision-boundary moments and amplify feedback there
System DOES -- bias task-graph branch selection from neural performance signal
System DOES -- maintain zero drift across all three feedback levels
System DOES NOT -- flip live action winners in current approach-dominant regime
System DOES NOT -- perform language reasoning or coding tasks
System DOES NOT -- use a token encoder or semantic representation layer
System DOES NOT -- learn updated representations from task outcomes
System DOES NOT -- generalize across task domains

Architecture layers (end of Day 45):

D38-41 -> prediction system, VSA representation, temporal behavioral control
D42 -> trajectory formation, outcome evaluation, boundary-level decision control
D43 -> structured environment dynamics, regime switching, preference/calibration
D44 -> deterministic cognitive stress-testing substrate (task cycles, probes)
D45 -> first bidirectional task-to-neural feedback bridge

Files modified:

ikigai/core/state.py -- all Day 43-45 fields (env, task loop, feedback bridges)
ikigai.py -- task loop injection, BG scoring feedback packs 1-2,
 task-graph branch bias Pack 3

Files created:

experiments/day43_pack{03..12}*.py (10 validation harnesses)
experiments/day44_pack{1..8}*.py (8 validation harnesses)
experiments/day45_pack{1..3}*.py (3 validation harnesses)

Design principles:

"Small signals + correct timing > large signals."
"Behavior emerges through constrained interaction, not brute-force scaling."
"The control architecture now exists; the language architecture does not yet exist."

Final principle:

"Days 43-45 did not create intelligence.
They created the infrastructure required for adaptive cognition.
The system is now stateful, deterministic, closed-loop,
and internally inspectable.
The system can now be influenced by its own task performance,
even if only weakly."

Status: Days 43-45 complete. Closed-loop behavioral substrate established. Ready for token encoder pivot.